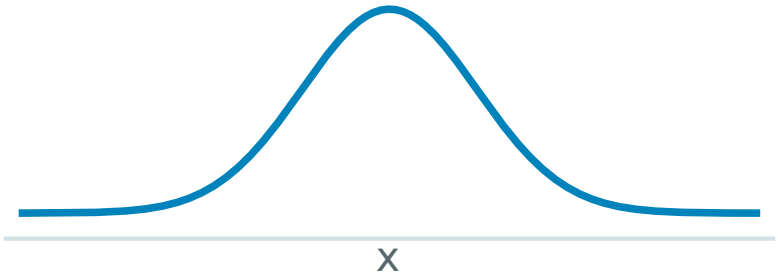
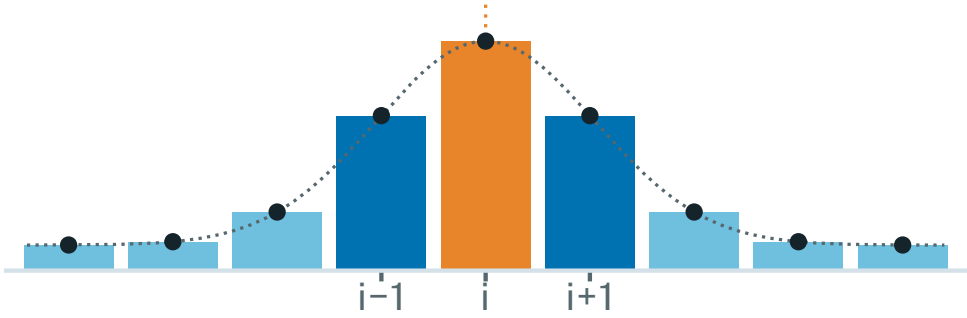
1 Continuous field $u(t_k, x)$

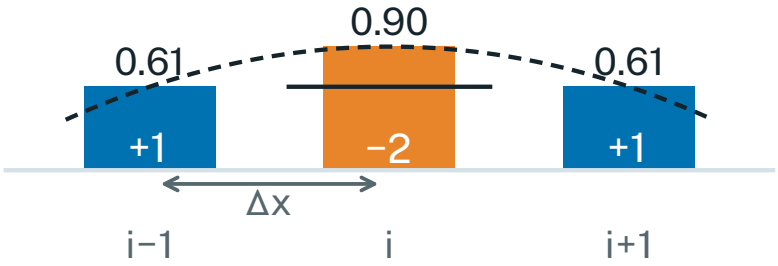


2 Grid values u_i^k (one per cell)

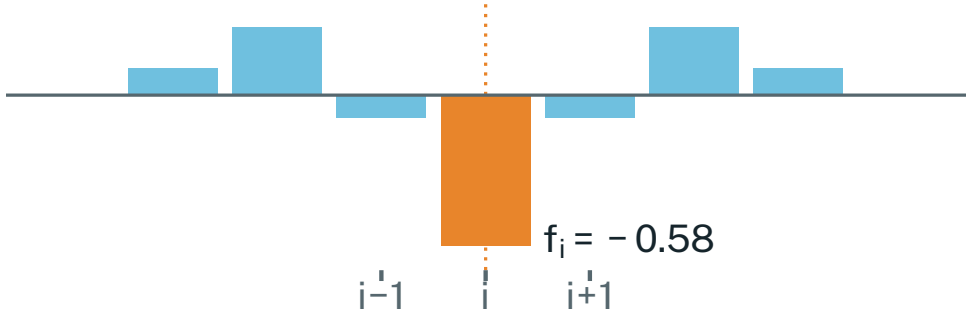


3 Stencil, here for cell i

$$f_i = (u_{i-1} - 2u_i + u_{i+1}) / \Delta x^2$$
$$= (0.61 - 2 \cdot 0.90 + 0.61) / 1^2 = -0.58$$

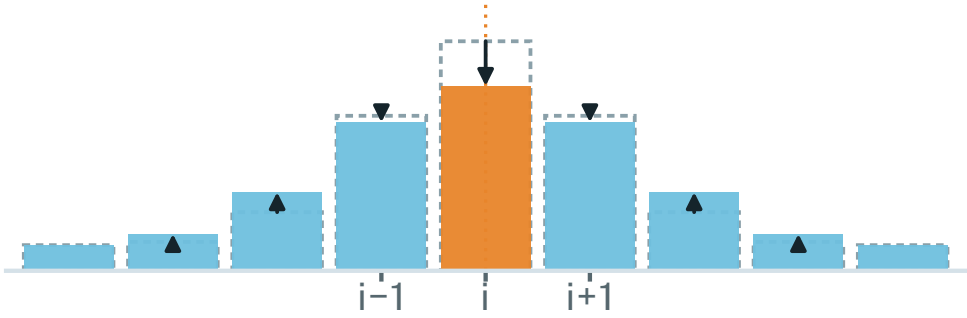


4 Rates f_i (the stencil at every cell)



5 Time step, here for cell i

$$u_i^{k+1} = u_i^k + \Delta t\, f_i$$
$$= 0.90 + 0.3 \cdot (-0.58) = 0.73$$

explicit Euler; RK4 or any scheme that evaluates f can replace it

6 Next state u_i^{k+1} (dashed: u_i^k)